

2023 AMC 10B Problem 3 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 3. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 3

Problem:

A $3 - 4 - 5$ right triangle is inscribed in circle A , and a $5 - 12 - 13$ right triangle is inscribed in circle B . What is the ratio of the area of circle A to the area of circle B ?

Answer Choices:

A. $\frac{9}{25}$

B. $\frac{1}{9}$

C. $\frac{1}{5}$

D. $\frac{25}{169}$

E. $\frac{4}{25}$

Solution:

The hypotenuse of a right triangle is a diameter of its circumscribed circle. Thus the diameter of circle A is 5, and the diameter of circle B is 13. Hence the requested ratio is

$$\frac{\pi \left(\frac{5}{2}\right)^2}{\pi \left(\frac{13}{2}\right)^2} = (\text{D}) \boxed{\frac{25}{169}}$$

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